

A Certified Admission Gate for Online Reconfiguration of Time-Sensitive Networks

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Preparatory work for the PhD thesis *Certified Dynamic Reconfiguration of Time-Sensitive Networks:
Online Admission Control under Worst-Case Determinism Guarantees*

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Outline

1. Problem
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1. Problem: Worst-Case Guarantees in a Changing Network

Context

Factories, vehicles and aircraft replace dedicated fieldbuses with **one Ethernet network** shared by control loops, sensors, video and best-effort traffic. For control traffic, a late message is a **failure**, not a slower service: what is needed is a **guaranteed worst case**.

- **TSN** (IEEE 802.1Q [1]) gives bounded latency, but only for a flow set **analysed offline**, with network calculus [2, 3] or a synthesised schedule [4].
- In operation, **flows are added while the network runs**; each new flow can delay flows already admitted, **even on other paths** that share a port.
- Fast **proposers** exist (heuristics, ILP/SMT, deep RL), but their outputs are **optimised, not proven**.

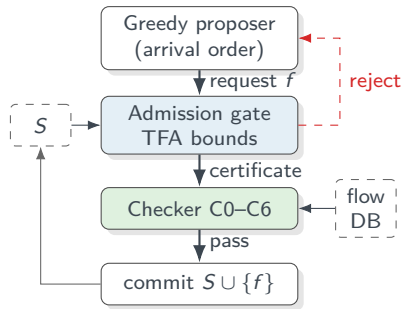
After this change, does every admitted flow still meet its deadline?

1. Problem: Our Approach and Contributions

A certified admission gate

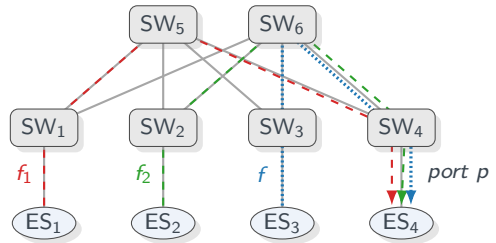
Between **any** proposer and the network, like a safety shield [5]: a change is committed only with a **certificate** re-verified by an **independent checker**. We develop:

- **C1.** A **TSN topology** with redundant core switches, feed-forward under ECMP.
- **C2.** An **admission gate**: a flow is accepted only if all deadlines still hold.
- **C3.** Incremental gate: affected ports only, **bit-identical** bounds.
- **C4.** Validation: **panco** [6], simulation, tests; reproducible.
- **C5.** Explicit limits of the guarantee.



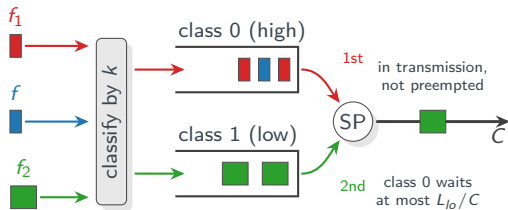
2. Model: Network, Flows and Scheduling

- **Network:** edge switches linked to every core switch, end stations on edge switches, full-duplex links of rate $C = 1 \text{ Gbit/s}$; each link direction is an output **port**; processing delay t_{proc} .
- **Routing:** fixed shortest paths, ECMP over core switches; port dependency graph checked **feed-forward**.
- **Flow** f : token bucket $\alpha_f(t) = b_f + r_f t$, max frame L_f , priority class k_f (0 highest), deadline $D_f^{\max}$.
- **Ports:** non-preemptive static priority (**NP-SP**), FIFO within a class.



ECMP routes f_1 (class 0) via SW₅, f_2 (class 1) and the new class-0 request f via SW₆; all share port p , so admitting f can delay both.

2. Model: Packet Handling and Delay Bounds at an NP-SP Port



Per-hop bound for class k at port p

Residual rate-latency service [3, 7]:

$$d_k = \frac{B_H + L_{lo} + B_k}{C - R_H} + t_{proc}, \quad R_H + R_k \leq C$$

Burst propagation and end-to-end bound (TFA)

$$b' = b + r_f d_k, \quad D_f = \sum_{h \in \mathcal{P}_f} d_{k_f}^{(h)} \leq D_f^{\max}$$

B_H, R_H : higher-priority bursts and rates; L_{lo} : largest lower-priority frame (non-preemptive blocking); B_k, R_k : class- k aggregates.

3. Admission Gate: Certificate and Independent Checker

Interface and invariant

On request f , analyse $S \cup \{f\}$; accept iff it is schedulable. **Invariant:** every flow in S meets its deadline. A rejection lists **every** violated flow, not only f .

Certificate

Per flow: path and, per hop, b_{in} , $(B_H, R_H, L_{lo}, B_k, R_k)$, d ; end-to-end bound; fingerprint of S before/after (XOR of SHA-256 flow hashes).

Checker (about fifty lines, no analysis code)

- C0 one witness per admitted flow, with its own parameters
- C1 each path is the fixed route
- C2 first input burst $\geq b_f$
- C3 next input burst $\geq b_{in} + r_f d$
- C4 claimed aggregates $\geq$ those recomputed from all witnesses at the port
- C5 $R_H < C$, $R_H + R_k \leq C$, $d \geq$ per-hop bound
- C6 end-to-end bound $\geq \sum d$ and $\leq$ deadline

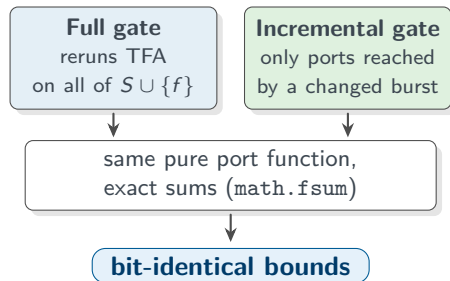
3. Admission Gate: Soundness and Incremental Computation

Proposition (soundness)

If a certificate passes C0–C6, every admitted flow meets its deadline under the model.

Why it holds

- The bound grows with B_H, R_H, L_{lo}, B_k .
- Induction over the **topological order of ports**: every witnessed burst over-approximates the true one (C2–C5).
- Summing the hops gives the deadline (C6).
- **C0 is necessary**: omitting a flow only *lowers* the others' aggregates; only the flow database reveals it.



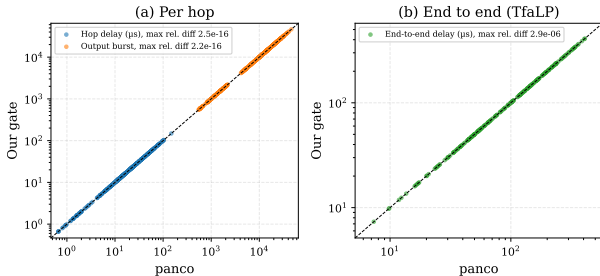
The incremental gate issues a **delta certificate** (changed flows only).

4. Results: Setup and Validation Against panco

Setup

Small	2 core $\times$ 4 edge switches, 56 ports
Large	4 core $\times$ 8 edge switches, 144 ports
Links	5 end stations per edge switch, ECMP, $C = 1$ Gbit/s
Class 0	64–256 B, deadline 0.1–0.5 ms
Class 1	500–1500 B, deadline 0.5–2 ms
Runs	greedy proposer, 1,000 requests $\times$ 10 seeds, both gates

Reference: panco [6], per-hop residual servers and end-to-end TFA as a linear program.



$$2.5 \times 10^{-16}$$

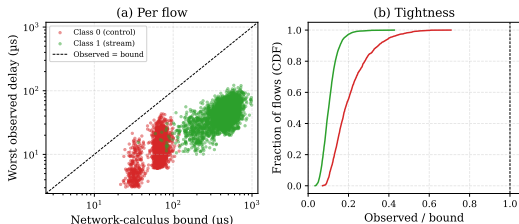
per hop, 1,119 values

$$2.9 \times 10^{-6}$$

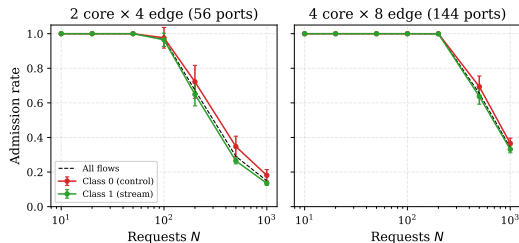
end to end, 242 values

max. relative difference (end to end: lp_solve prints 6 digits)

4. Results: Simulation and Admission Rate

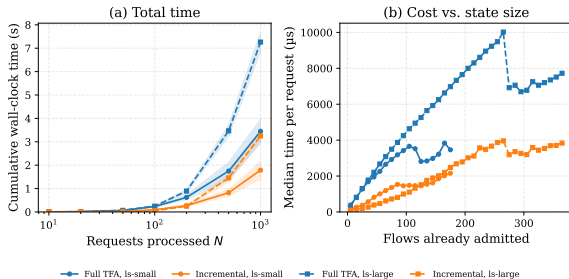


0 violations over 4,915 admitted flows; median observed/bound 0.19 (class 0), 0.10 (class 1). A sanity check, not a tightness measure.



All requests accepted up to 50 (small) and 200 (large); 150 ± 14 and 342 ± 22 admitted of 1,000; class 0 accepted slightly more often.

4. Results: Run Time of the Full and Incremental Gates



Topology	Gate	Median μ s	Speed-up
Small	Full	3097	1.97×
	Incremental	1617	
Large	Full	7317	2.24×
	Incremental	3331	

- Identical verdicts and **bit-identical** bounds.
- Gain grows with size: a request re-evaluates 24 of 56 ports (small) but 49 of 144 (large), median.
- 18 unit tests, incl. tampered certificates (C5 and C0 reject them).

5. Limitations

Model

Single link rate, fixed ECMP routing, feed-forward (no rings), strict NP-SP; no CBS/TAS shapers, preemption or clock effects; Ethernet overheads folded into L_f, b_f, r_f .

Analysis

TFA is sound but pessimistic compared with PMOO or LP-based analyses [8, 7]: fewer admissions, never an unsafe one.

Certification

Proof sketch on paper, not mechanised; floating-point checker with 10^{-9} tolerance; the certificate attests to the model, not to the deployed switches.

Evaluation

Additions only; two-tier topologies of at most 144 ports, two classes; simulator does not force the worst case; only a greedy proposer (arrival order), no ILP or DRL yet.

6. Conclusion

Contribution

- **TSN topology** with redundant core switches, feed-forward under ECMP.
- **Safe traffic allocation**: a flow is admitted only if every deadline still holds.
- **Certificate** per decision, re-checked by a 50-line checker (C0–C6).
- **Any proposer**, even a learned one, cannot commit an unsafe allocation.
- **Incremental gate**: 1.6–3.3 ms, bit-identical bounds, $2.0\text{--}2.2\times$ faster.
- **Validated**: matches panco; 0 violations over 4,915 flows.

Future directions

- **Capacity**: tighter, still checkable analyses than TFA.
- **Coverage**: CBS and TAS shapers, frame preemption.
- **Trust**: mechanised proof, exact-arithmetic checker.
- **Operations**: flow removal and batched changes.
- **Proposers**: the gate as a shield behind heuristic, ILP and DRL.

Thank you for your attention!

Questions?

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